

Young Children's Understanding of Posterior Probabilities

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Background

- Reasoning about **posterior probabilities** involves first making a prior probabilistic judgment and then updating that judgment based on new information
- Limited research has examined how and when children develop the ability to reason about posterior probabilities
- Giroto & Gonzalez (2008) found that children do not reliably reason about posterior probabilities **until around age five**
- We adapted their task into a new block game, removing indirect elements (e.g., character ownership and betting) to allow for **straightforward assessments** of children's understanding of posterior probabilities

Key Reference:

Giroto, V., & Gonzalez, M. (2008). Children's understanding of posterior probability. *Cognition*, 106(1), 325–344.

Method

- Experiment 1:** 5- and 6-year-old children (N = 48) played a probability updating block game (see center section for more details)
- Experiment 2:** 4-year-old children (N = 24) played the same game with the same within-subject, repeated-measure design
- Within-subject, repeated-measure design:** each child received eight test trials with varying colors and shapes, organized in four pairs of trials (prior & posterior)

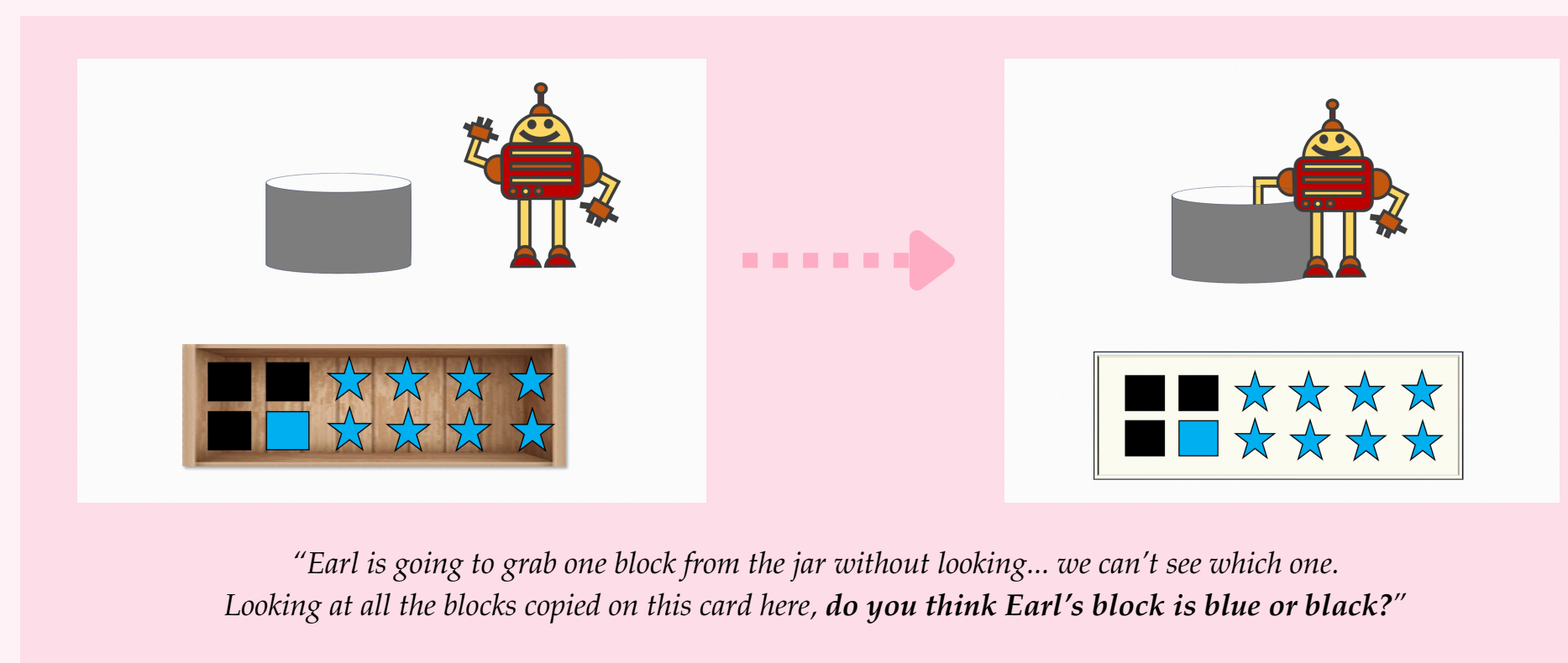
Hypothesis

If young children are able to understand posterior probabilities, then they should be able to:

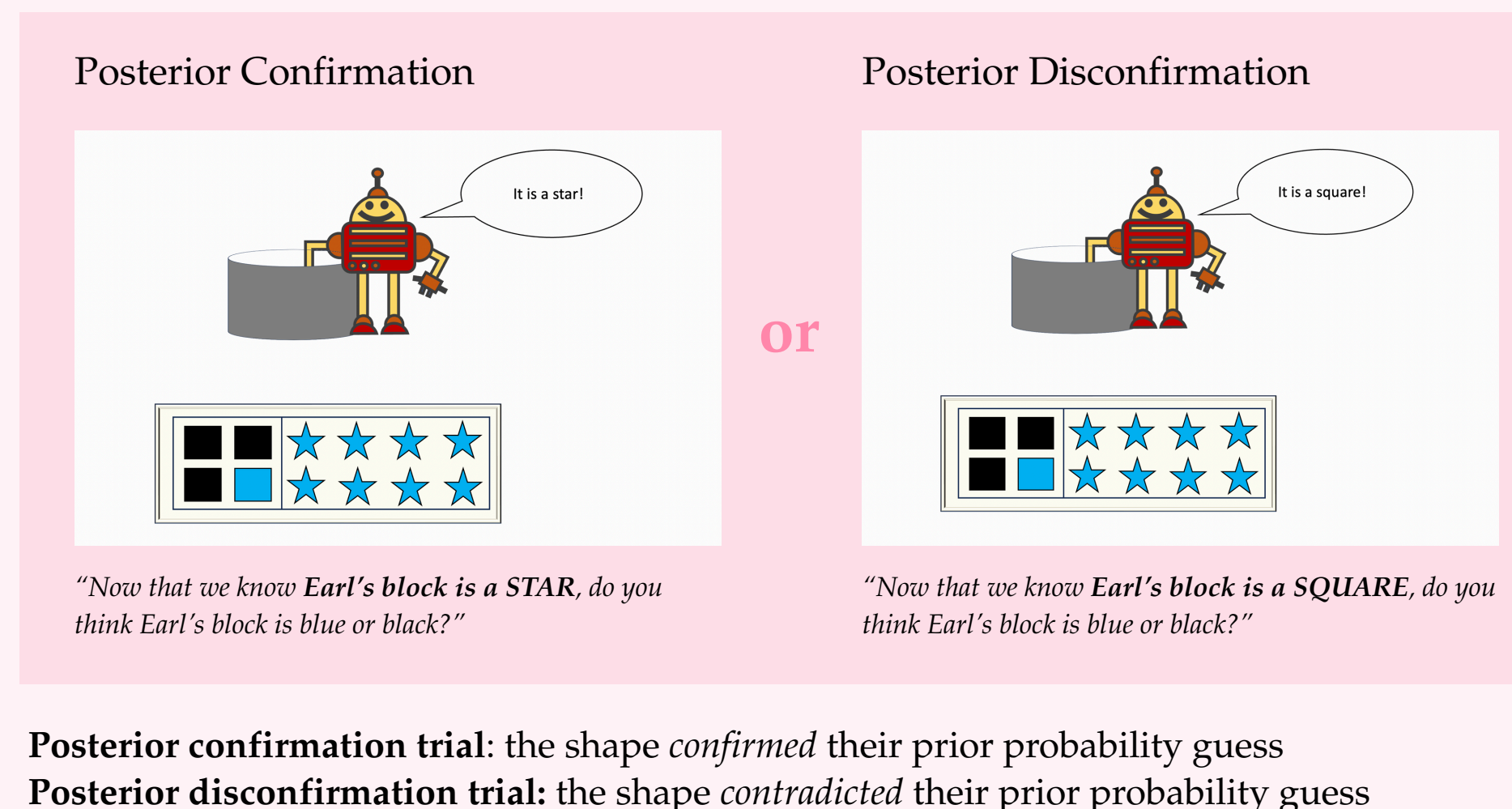
- make accurate **prior probability guesses** based on the overall distribution of block colors; and
- accurately **update their prior probability guesses** when provided with new information about the block's shape

Success at this probability updating task suggests that children can reason about posterior probabilities

Prior Trial: no shape information; should guess the more probable color



Posterior Trial: shape revealed

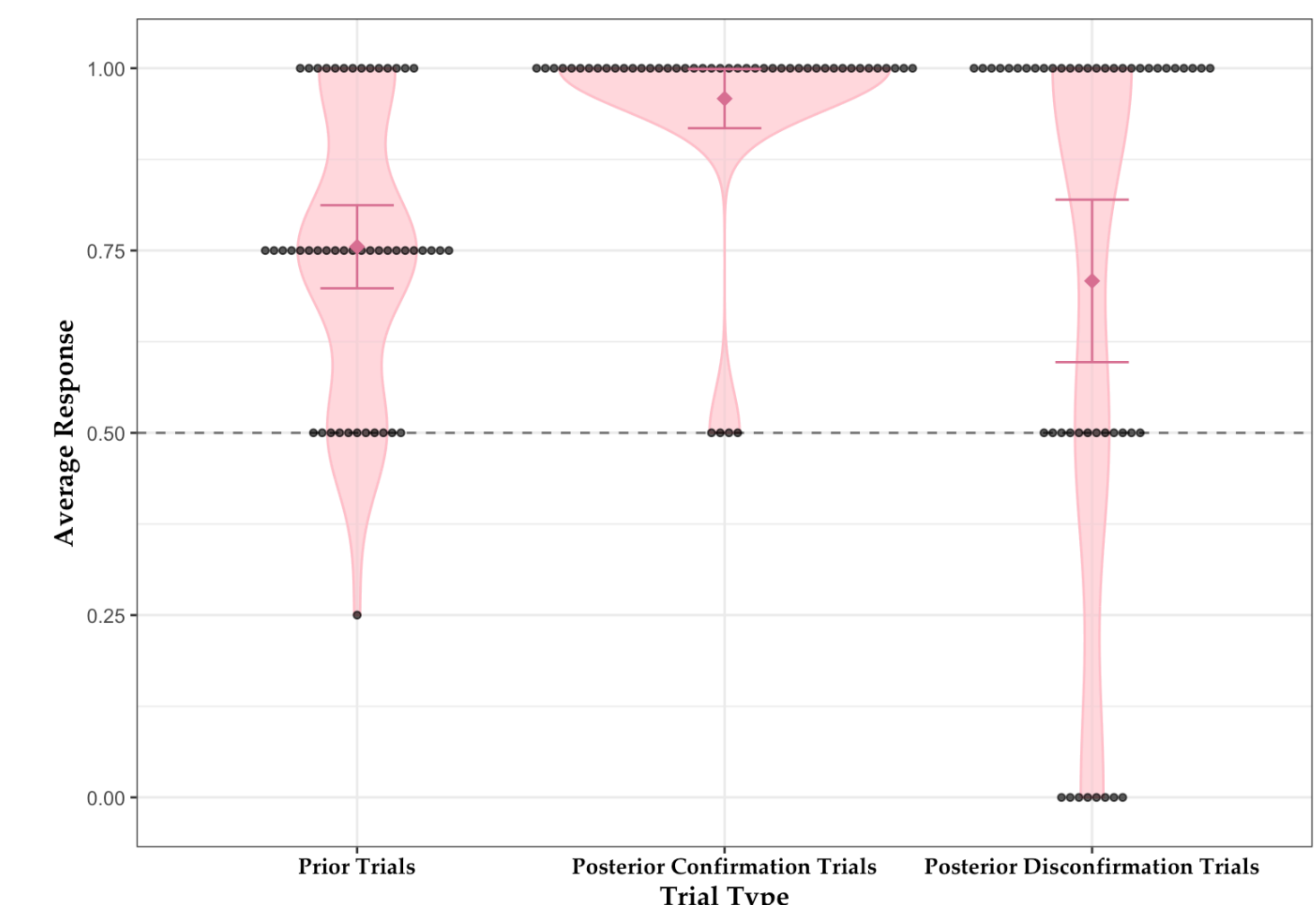


Discussion

- 5- and 6-year-olds show **robust ability** to reason about posterior probabilities
- 4-year-olds show **emerging ability** to reason about posterior probabilities
- Ability to reason about posterior probabilities **may have emerged earlier in life** than previously thought; future studies should examine if even younger children can do so

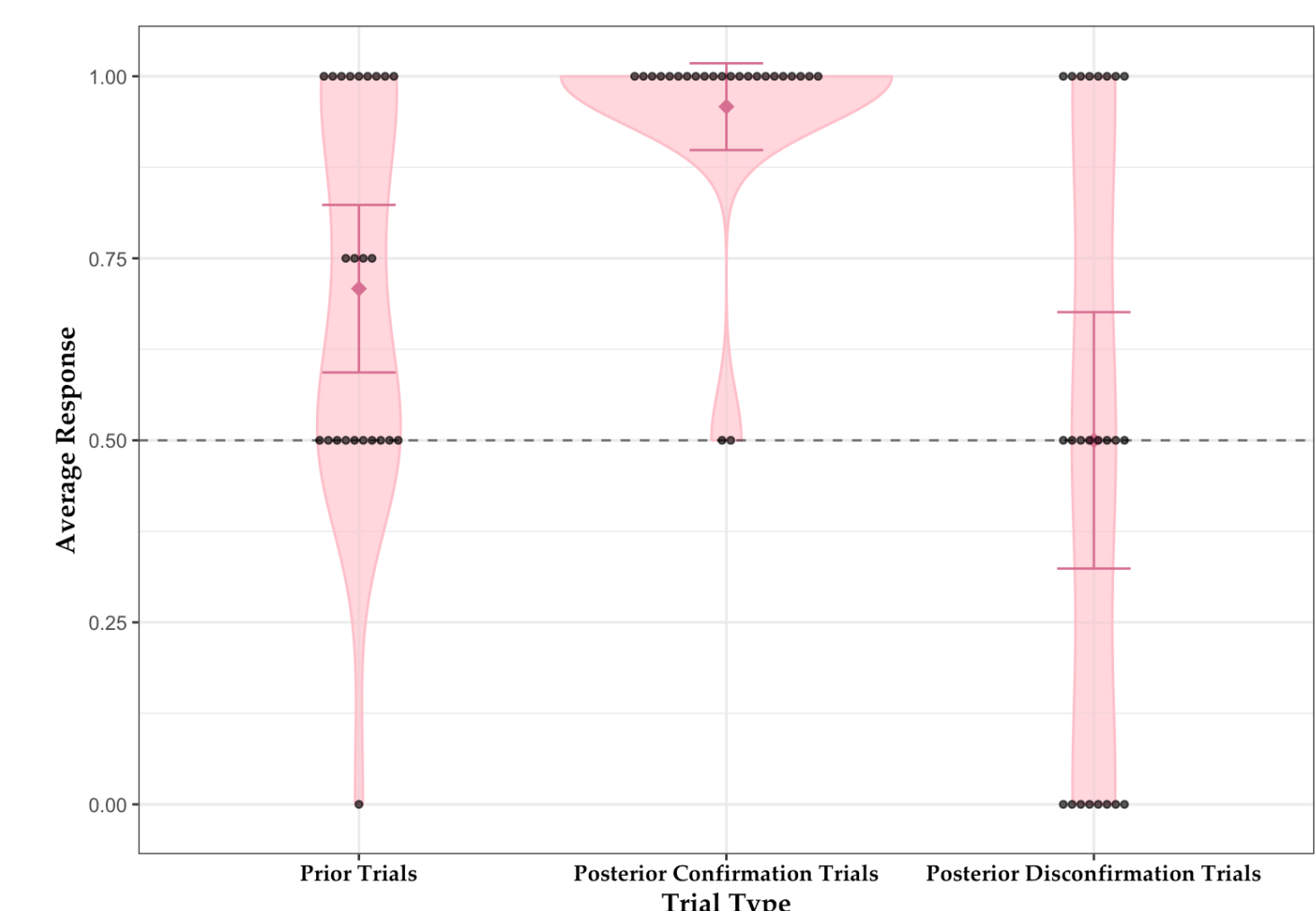
Key Findings

Experiment 1: 5- and 6-year-olds



- 5- and 6-year-olds performed above chance in **all trial types** ($p < 0.001$);
- They accurately updated their prior probability to a posterior probability regardless of the type of evidence (confirmation vs. disconfirmation) provided

Experiment 2: 4-year-olds



- 4-year-olds performed above chance in the **prior trials** ($p = 0.003$) and the **posterior confirmation trials** ($p < 0.001$), but not in the posterior disconfirmation trials ($p = 0.51$);
- They accurately updated their prior probability guess when presented with confirming, but not contradictory, evidence